Fixopolis
Prototype Design Document

The working spec for the Part II interactive prototype: what we build, why each part exists, and how it's put together.

Unit INFO3315 Human-Computer Interaction **Version** 1.7 (draft), 19 Oct 2026 **Builds on** `civic-app-prototype.html` (v1)

Due Demo + Poster, Sun 25 Oct

> **How to read this.** Decided marks choices the group has made. Proposed marks suggestions that need group sign-off before they're built. Heuristic references (H1 – H10) are Nielsen's 10, numbered as in the Week 8 worksheet.

1 Purpose and scope

We're building a high-fidelity, clickable prototype that lets participants complete **three test tasks** (Brief 6a) and records their performance automatically (Brief 6b). It isn't a full product (Brief 6c). Every screen and control exists to support those tasks or a recorded evaluation finding. No extra features: the brief gives no marks for them.

In scope

- T1: report a problem (photo or written)
- T2: check the status of your own report
- T3: find someone else's report and bump it
- Login, home, profile, help (supporting screens)
- Built-in test mode: guided tasks, post-test survey, results saved to CSV automatically

Out of scope

- Admin / civic officer side (PACT Task 3)
- Local news, notifications, real camera/GPS, backend
- Severity, ETA prediction, community stats (in Part I but not needed for the test tasks; listed as future work)

Format Decided: one self-contained HTML file, no build step, so anyone in the group can open and edit it. Shared via a private link for the demo submission. The tutor accepted a non-Figma prototype.

2 Research basis

Every feature traces back to Part I. Sources: the survey (n = 42), five interviews, the competitor analysis and the PACT analysis.

Finding	Source	Design response
67% have never reported an issue. Non-reporters don't know where to report or doubt it's worth the effort.	Survey; persona table	One obvious entry point on Home; report form reached in 2 taps; no account set-up in the flow.
61.9% don't understand what their council is doing about issues. Satisfaction with reporting is closely linked to perceived transparency.	Survey (Spearman analysis)	Status badge on every report; dated status timeline on Report Details.
Reporters doubt anything will happen: "visible follow-through" is their core need.	Persona table; Fiona's, Artem's interviewees	Updates timeline; "Get updates" toggle; a report stalled "In progress" in the dummy data so T2 tests reading a stalled status.
Duplicate reports could signal severity to admins. People want to see other reports being fixed.	Artem's, Olivia's interviewees	**All reports** tab with bumping (T3); bump count shown on each report.
Concerns about personal data and revealing your whereabouts.	Satria's, Artem's interviewees	"Include location" toggle with manual alternative; verified login; no public names on other people's reports.
SeeClickFix's anonymous reports were misused; Find It, Fix It caps submissions and is limited to one service area.	Competitor analysis	Verified (biometric/PIN) account; no submission cap; location from GPS, not a fixed council area (Universality criterion).
A FixMyStreet reviewer wanted to see whether the council actually acts.	Competitor analysis	Status visible from the report list, without opening each report.

Finding	Source	Design response
Most-mentioned issues: roads/potholes (19), rubbish (7), footpaths (6), utilities/infrastructure (6).	Open survey question	Category set and the mix of dummy data (§7).

3 Evaluation findings and change log (v1 → v2)

v1 = `civic-app-prototype.html` and the Excalidraw wireframes. Each change is tagged with its source so the final report can document it (Brief 6f). Sources: Think-aloud (TA), Heuristic Evaluations Workbook (HE), group decision (G). The cognitive walkthrough produced no actionable issues.

ID	Finding	Heuristic	Change in v2	Status
G1	Third task needed (Brief 6a)	n/a	"Current issues" screen with **Your reports / All reports** tabs; bumping from All reports. Replaces "Check previous reports" and "Find updates on local issues".	Decided
G2	Every v1 report is saved as "Pothole"	n/a	Category picker added to the report form (PACT Task 1; bumping relies on categories too).	Decided
G3	Council officials are stakeholders (PACT); app needs a separate sign-in for them	n/a	"Sign in as a council official" option on Login, opening a staff ID sign-in that always rejects the credentials (§10).	Decided
TA1	Use icons and interface metaphors, not just words	H2 H6	Home becomes an icon tile grid; category icons; map-pin metaphor for location.	Decided
TA2	Profile button top right, as standard	H4	Avatar button top right of Home → Profile screen.	Decided
TA3	"Implement more buttons": icon buttons, not just text	H2 H6	Every main action gets an icon + label (Home tiles, report type, Bump, Help, Profile).	Decided
HE1	Report list shows only a date, not status	H1	Status badge (colour + label) on every report row.	Decided
HE2	No confirmation of login	H1	"Signed in" toast on Home.	Decided
HE3	Splash has no loading indicator	H1	Spinner on splash.	Decided
HE4	Users may not have Touch ID	H2	Login offers Face ID, Touch ID and PIN.	Decided
HE5	Can't edit or delete a submitted report	H3	Delete (with confirmation) on your own reports. Edit is nice-to-have.	Decided
HE6	Back navigation inconsistent	H4	One `NavBar` component: "‹ Back" on every screen below Home.	Decided
HE7	No handling for missing location	H5	Can't proceed without a location; "Location unavailable" banner with a "Choose on map" button.	Decided
HE8	Report items have no visual cues	H6	Category icon + status badge on every row.	Decided
HE9	No "stay signed in" option	H7	"Stay signed in" toggle on Profile (simulated).	Decided
HE10	Report Details sections separated only by text weight	H8	Separate cards with section labels and whitespace.	Decided
HE11	No error states	H9	Inline red error + fix action for: no photo, no category, no location.	Decided
HE12	No help section	H10	"?" icon button top left of Home, opposite the profile avatar → Help screen: what the app is and how to use it.	Decided

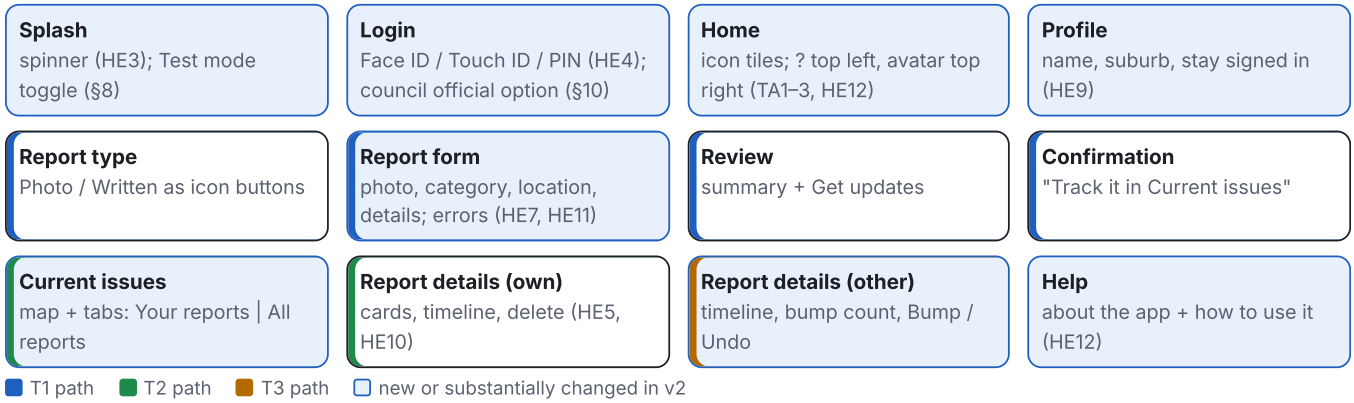
4 Test tasks

These definitions drive logging and automatic completion detection. Participant-facing wording goes in the separate task sheet (what to do, never how; max 3 tasks).

Task	User goal	Start	Optimal path	Min. taps	Success (auto-detected)
T1	Report a pothole outside your home	Home	Report a problem → Photo report → take photo → pick "Road / pothole" → Next → Submit	6	A new report exists with category = road and a location
T2	Find out what's happening with your footpath report	Home	Current issues → Your reports → "Broken footpath"	2	Details for report `R-102` are opened
T3	Support a neighbour's report of a broken streetlight	Home	Current issues → All reports → "Broken streetlight" → Bump	4	`R-205` is bumped by the current user

Measures (Lecture 8): completion (yes/no), time on task, taps, **dead taps** (taps that hit nothing interactive; used as an error count), back presses, and path efficiency = min. taps ÷ actual taps, plus a post-test survey. Test mode collects all of these automatically (§8).

5 Screens and flow



Navigation: Splash → Login → Home. Home branches to Report type, Current issues, Help, and Profile. Current issues → Report details (own or other, depending on the tab). Confirmation → Home (stack reset).

Screen notes

Screen	Key behaviour
Home	Top bar: **Help** (?) top left, **Profile** avatar top right. 3 large icon tiles: **Report a problem** (megaphone), **Current issues** (map pin), **View local news** (newspaper; placeholder, "not part of this prototype").
Report form	Photo slot (tap to "capture" a stock pothole image; Retake). Category chips with icons. Location row: "Current location ✓" or a red error + "Choose on map" when the toggle is off. Optional details. **Next** checks the form and shows inline errors rather than a blocking alert. **Written report** Decided : a working path using the same form without the photo slot.
Current issues	Segmented control: **Your reports** | **All reports**. Map shows pins for the active tab. Rows: category icon, title, street, status badge; All reports rows also show the bump count.
Details (other)	Bump button: "Bump (6)" → "Bumped ✓ · Undo (7)", plus a toast. One bump per user. No bump button on your own reports.
Help	Two short sections: **About** (what the app does, who receives reports) and **How to use** (report a problem, track your reports, bump a neighbour's report), each with its icon.
Details (own)	Summary card, Updates card (timeline, newest first), Delete button → confirmation dialog (H5).

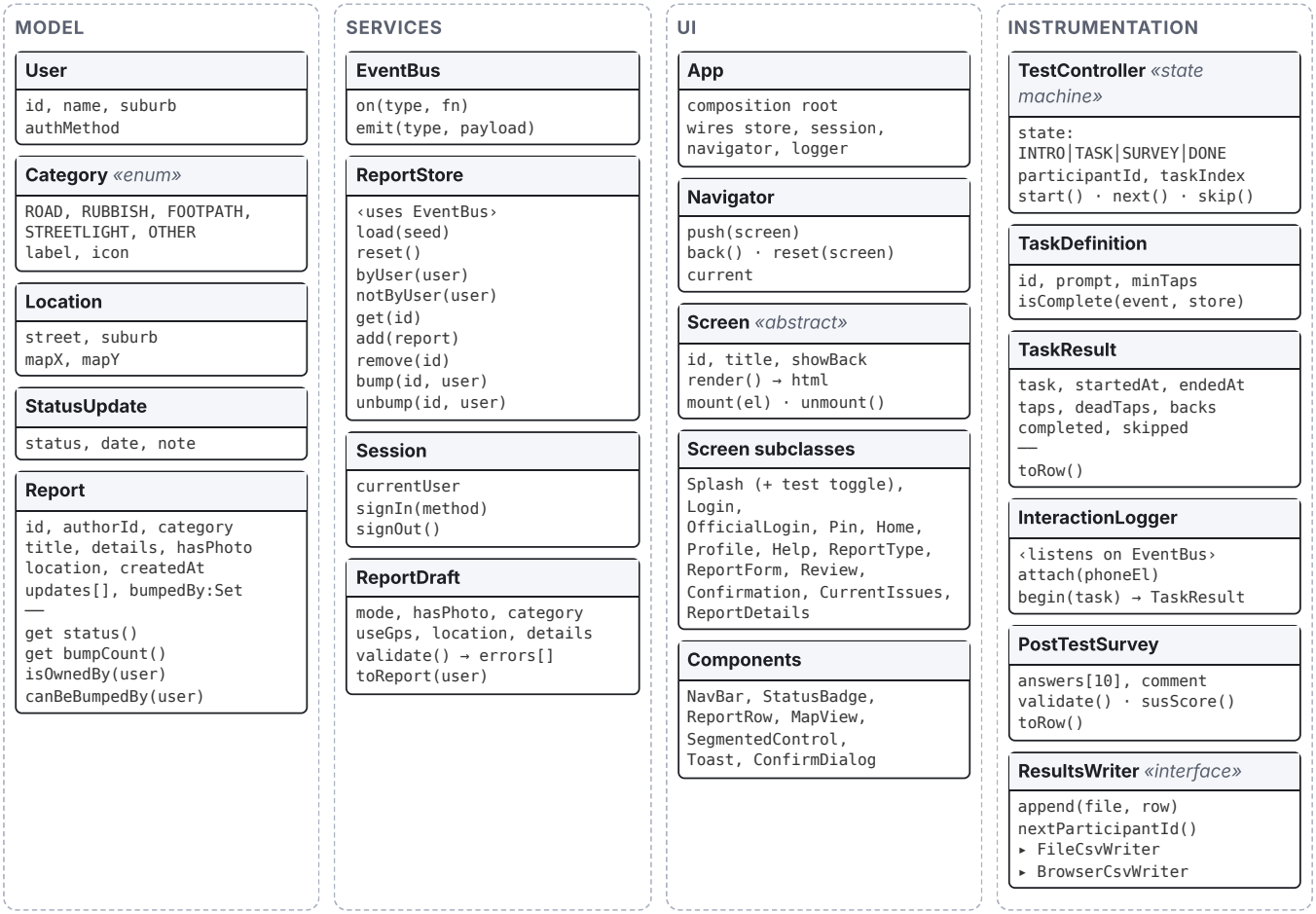
Visual and interaction rules

- Follow iOS conventions (platform consistency, H4): large title on Home, "‹ Back" nav bars, 44 pt minimum tap targets.
- Status is shown by **colour + text label**, never colour alone: Received (grey), Assigned (blue), In progress (amber), Fixed (green).

- Errors: red text under the field, saying what's wrong and how to fix it (H9).
- Every action gives immediate feedback: toast, badge change or screen transition (H1).

6 Architecture (OOP)

Vanilla JavaScript ES classes in one file, split into four layers. UI classes never change data directly: they call the store, and the store notifies listeners.



Pattern	Where	Why
Encapsulation	`Report` computes `status` and `bumpCount`; rules like "can't bump your own report" live in `canBeBumpedBy()`	Business rules are in one place, not spread across screens.
Inheritance + template method	`Screen` defines the lifecycle (`render → mount → unmount`); subclasses override `render()`	Consistent nav bar and back behaviour on every screen (HE6).
Repository	`ReportStore` is the only owner of report data; `reset()` reloads the seed	Identical starting state for every participant.
Observer	`EventBus`: store emits `report:added/bumped/removed`; navigator emits `screen:shown`; logger listens	Logging stays separate from UI code and is easy to switch off.
State	`TestController` moves Intro → Task 1–3 → Survey → Done; each state decides which popup shows	The test sequence can't skip or repeat a step by accident.
Polymorphism (strategy)	`ResultsWriter` interface; `FileCsvWriter` (main) and `BrowserCsvWriter` (safety net) behind one interface	Test code doesn't care where results are stored (§8).
Composition	`App` builds and injects dependencies; screens are composed from components	Each piece can be tested and swapped on its own.

7 Dummy data

Fictional suburb **Elmwood**, current user **Sam Taylor** (no real names or addresses). The category mix follows the survey's open responses. Dates are fixed so every participant sees the same state.

ID	Owner	Category	Title · location	Status history	Bumps	Used by
R-101	Sam	Road	Pothole · Station Rd	Received 3 Sep → Assigned 5 Sep → Fixed 12 Sep	3	
R-102	Sam	Footpath	Broken footpath · Hill St	Received 18 Sep → Assigned 22 Sep → In progress 25 Sep (stalled)	2	T2 target
R-103	Sam	Rubbish	Dumped rubbish · Park Ln	Received 21 Aug → Assigned 22 Aug → Fixed 24 Aug	0	
R-201	Other	Road	Pothole · Elm Ave	Received 2 Oct → Assigned 6 Oct	9	
R-202	Other	Road	Cracked road surface · Mill St	Received 28 Sep	4	
R-203	Other	Rubbish	Overflowing bins · Elmwood Station	Received 9 Oct → Fixed 11 Oct	12	
R-204	Other	Footpath	Raised paver · School Rd	Received 1 Oct → In progress 8 Oct	5	
R-205	Other	Streetlight	Broken streetlight · Wattle St	Received 10 Oct → Assigned 13 Oct	6	T3 target
R-206	Other	Road	Faded crossing lines · Hill St	Received 14 Oct	1	T3 distractor (same street as R-102)
R-207	Other	Other	Fallen tree branch · Elm Park	Received 15 Oct	2	

Kept as one `SEED` constant at the top of the file. Example shape:

```
{ id:"R-205", authorId:"u2", category:"STREETLIGHT", title:"Broken streetlight",
  location:{ street:"Wattle St", suburb:"Elmwood", mapX:236, mapY:78 },
  details:"Light out for a week, very dark at night.", hasPhoto:true, createdAt:"2026-10-10",
  updates:[ {status:"RECEIVED", date:"2026-10-10"}, {status:"ASSIGNED", date:"2026-10-13", note:"Lighting team"} ],
  bumpedBy:["u3","u4","u5","u6","u7","u8"] }
```

8 Test mode and results

Testing is built into the prototype: no separate facilitator panel, stopwatch or manual note-taking for the numbers. Decided

1 · Splash
facilitator switches **Test mode** on

2 · Intro popup
"You'll do 3 tasks", lists them; Start

3 · Task 1
popup with task → Begin → timed

4 · Task 2
"Done! Next: ..." popup → Begin

5 · Task 3
"Done! Last one: ..." popup → Begin

6 · Post-test survey
popup after Task 3; Submit

7 · Thank you
results saved

8 · Reset
back to Splash for the next participant

Behaviour

- **Toggle off**: the normal prototype (Splash → Login → Home), for the class demo. **Toggle on**: a test run starts.
- **On start**: the next participant ID is assigned (P1, P2, P3 … incrementing), the data resets to `SEED`, the user is signed in automatically (login isn't a test task), and the app opens on Home.
- **Each task** starts from Home when the participant taps Begin, and the timer starts then. Completion is detected automatically by `TaskDefinition.isComplete()`; the result row is saved straight away, and the next task's popup appears.
- **Task reminder** Decided: a small "Task 2 of 3" pill at the top. Tapping it shows the task again, with a "Skip task" option (saved as skipped). This lets stuck participants leave a task (H3) and lets them re-read it without asking the facilitator.
- **Task wording** comes from the task sheet and is kept in one `TASKS` constant, so the prototype and the submitted sheet always match.

- **Ethics**: participant IDs only, never names; consent form signed before the run (Lecture 8).

Post-test survey

Decided The SUS (System Usability Scale): the 10 standard statements, each rated 1–5, plus one optional comment box ("Anything confusing or missing?"). All 10 must be answered before Submit; a missed item shows an inline error (H9). Score = (sum of odd items − 5 + 25 − sum of even items) × 2.5, giving 0–100.

Results files

File	Written	Columns
`results_tasks.csv`	One row as each task ends (3 per participant)	`participant_id, run_started, task, completed, skipped, time_s, taps, dead_taps, backs, min_taps, efficiency`
`results_survey.csv`	One row when the survey is submitted	`participant_id, submitted_at, sus_q1 … sus_q10, sus_score, comment`

Both files share `participant_id`, so the task results and survey answers can be joined for the Brief 6e/6f analysis.

> **How the CSVs update automatically** Decided : tests run only on a laptop, in **Chrome or Edge**. For security, a web page can't write to your files without permission, so the first time Test mode is switched on, the facilitator chooses a results folder once. From then on, rows are added to both CSVs automatically, and participant IDs continue from the highest ID already in the file. The browser may ask to confirm access once per session. (`FileCsvWriter` , using the File System Access API.)
>
> - **Use Chrome or Edge, not Safari or Firefox**: those two browsers don't support this. On a Mac, use Chrome.
> - **Safety net**: rows are also kept in browser storage, and a "Download results" button on the Splash screen saves both CSVs if folder access is ever refused. (`BrowserCsvWriter` .)

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9 Build plan

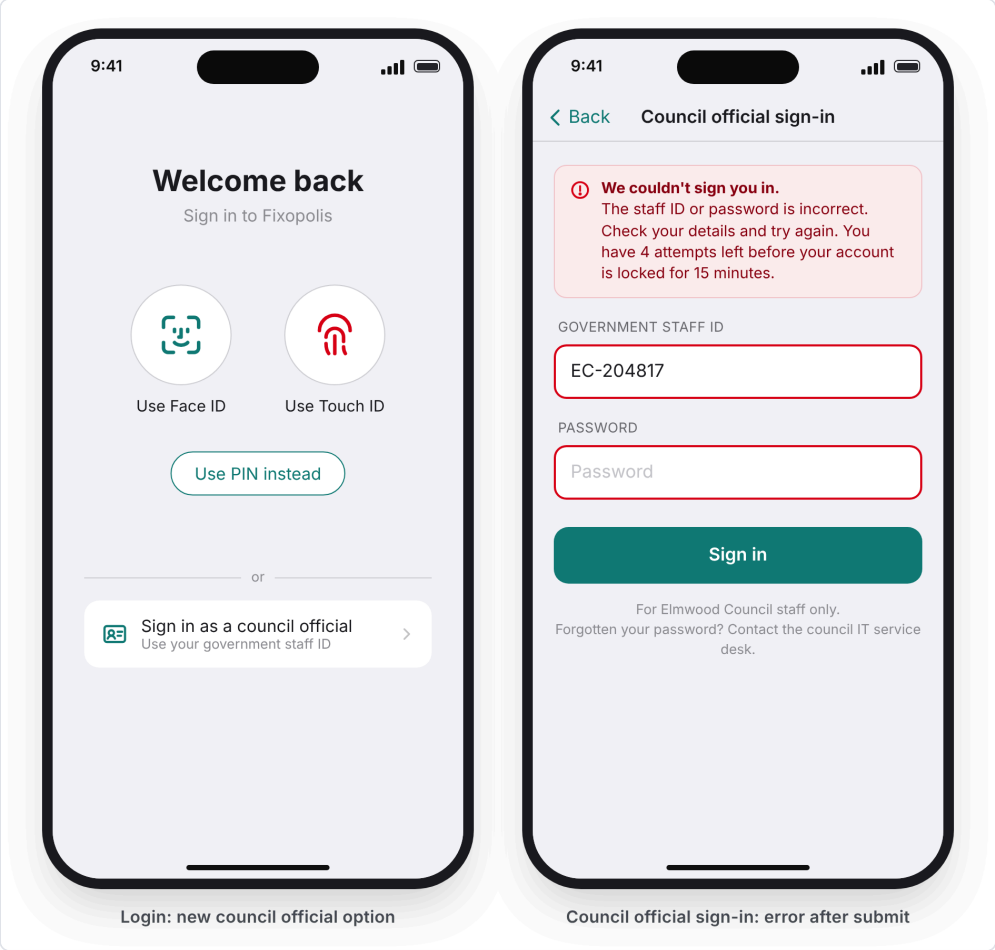
#	Step	Done when
1	Group signs off the Proposed items in §3 and §4	Tags updated in this doc
2	Model + ReportStore + SEED	Store loads, resets, bumps correctly
3	Screen base class, Navigator, NavBar; port v1 screens	v1 flows work on the new architecture
4	v2 changes: Home tiles, Profile, Help, report form, Current issues tabs, bumping, details cards	T1–T3 completable by the optimal path
5	Error states and confirmations (HE5, HE7, HE11)	Each error shows a message and a fix
6	Test mode: controller, task popups, logger, post-test survey, CSV writers	A dry-run participant produces correct rows in both CSVs
7	Internal pilot: group cognitive walkthrough of T3 (new flow); fix issues	No blocking issues
8	Publish link; export prototype PDF; write the task sheet	Demo submission ready by Sun 25 Oct
9	Test with 3 people outside the group (Brief 6d); log problems; prioritise fixes	Data and issue list ready for the final report
10	Restyle the prototype in the chosen visual direction (§11); add council official sign-in (§10)	All screens use the new tokens; T1–T3 still pass

Sign-off

Items in §1–§9 approved by the group (16 Oct 2026). Still to decide: the visual direction (§11).

10 Council official sign-in (non-functional)

Like websites that offer "log in as a student" or "log in as an educator", Login gets a second route for council staff. Decided



Login: new council official option

Council official sign-in: error after submit

Why	Civic officers, public officials and councils are PACT stakeholders (PACT Task 3, the admin side). The option shows the app serves both sides of the Reciprocity criterion.
Scope	Not usable in the prototype: the test persona is a typical resident and the admin side is out of scope (§1). Not part of any test task.
Entry	Login → "Sign in as a council official · Use your government staff ID" (below an "or" divider).
Form	Government staff ID, Password, Sign in. Footer: "For Elmwood Council staff only. Forgotten your password? Contact the council IT service desk."
On Sign in	Button shows "Checking…" for about 1 s, then the error: "We couldn't sign you in. The staff ID or password is incorrect. Check your details and try again. You have 4 attempts left before your account is locked for 15 minutes." Both fields outlined red, password cleared, attempts count down each try.
At 0 attempts	"Too many attempts. Your account is locked for 15 minutes. Contact the council IT service desk." Sign in is disabled.
Empty fields	Inline errors: "Enter your staff ID" / "Enter your password" (H9).
Data	Nothing is sent or stored. The council and ID format (EC-######) are fictional; no real government branding.
Code	`OfficialLoginScreen extends Screen` ; attempts held on the screen instance.

11 Visual design directions (choose one)

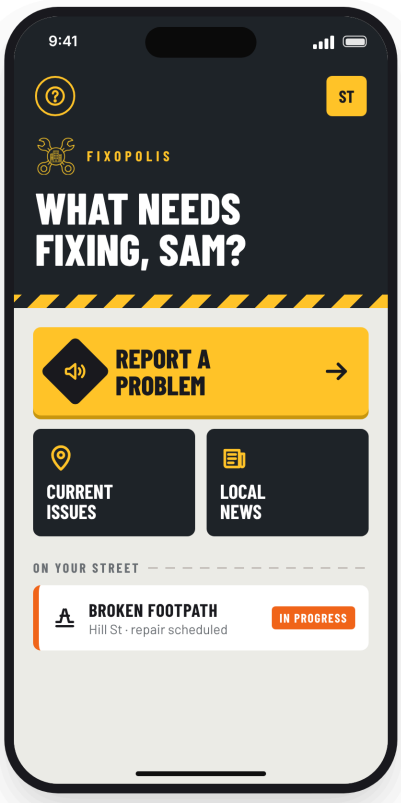
Peer feedback: v2 looks generic, like stock phone settings. Five directions follow, each shown on the same three screens: **Home**, **Report a problem** and **Report details** (the bump screen). Only the visual layer changes: screens, flows, tasks, heuristic fixes and test mode stay as specified in §3–§8. The logo is recoloured to suit each direction.

Direction	Character	Strongest link to research	Build effort vs v2
A Works Crew	Bold, high-contrast, roadworks	Potholes most-mentioned issue	Medium
B Wayfinding	Clear, civic, map-led	Location; transparency of next steps	High
C Council Docket	Official, precise, paper	Accountability and transparency gap	Medium
D Neighbourhood	Warm, friendly, social	Low self-efficacy; seeing others' fixes	Medium
E Blueprint	Technical, calm, on-brand	Consistency with logo; H2 metaphor	Medium

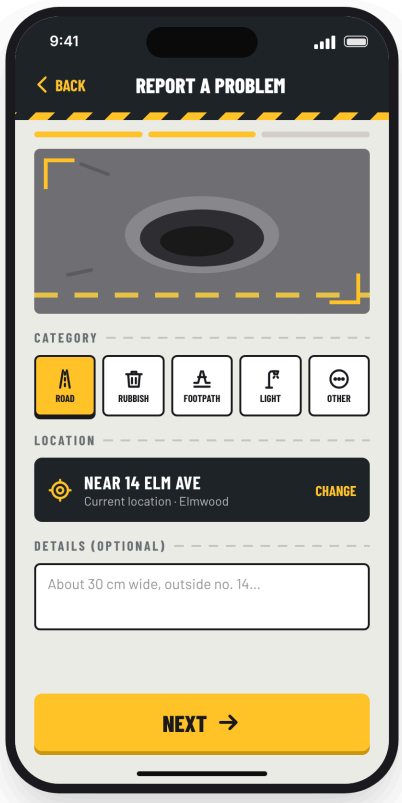
Directions can be mixed (for example, B's transit-line progress inside D). Once the group picks one, it becomes §5's visual rules and build step 10.

Direction A: Works Crew

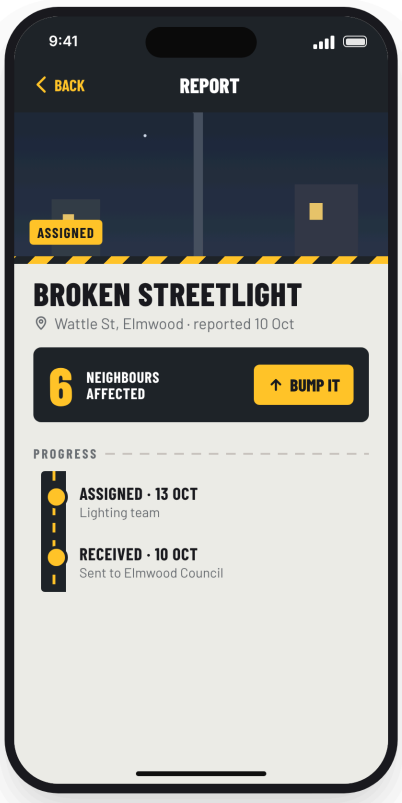
The look of council road crews: asphalt, hi-vis yellow, hazard stripes and road-sign lettering.



Home



Report a problem



Report details (bump)

Why it fits Roads and potholes were the most-mentioned issue (19 of 41 open responses). The styling says "the council is on it". Progress is drawn as a road with a dashed centre line, and categories sit on square sign plates.

Type Barlow Condensed (headings, road-sign style) · Barlow (body)

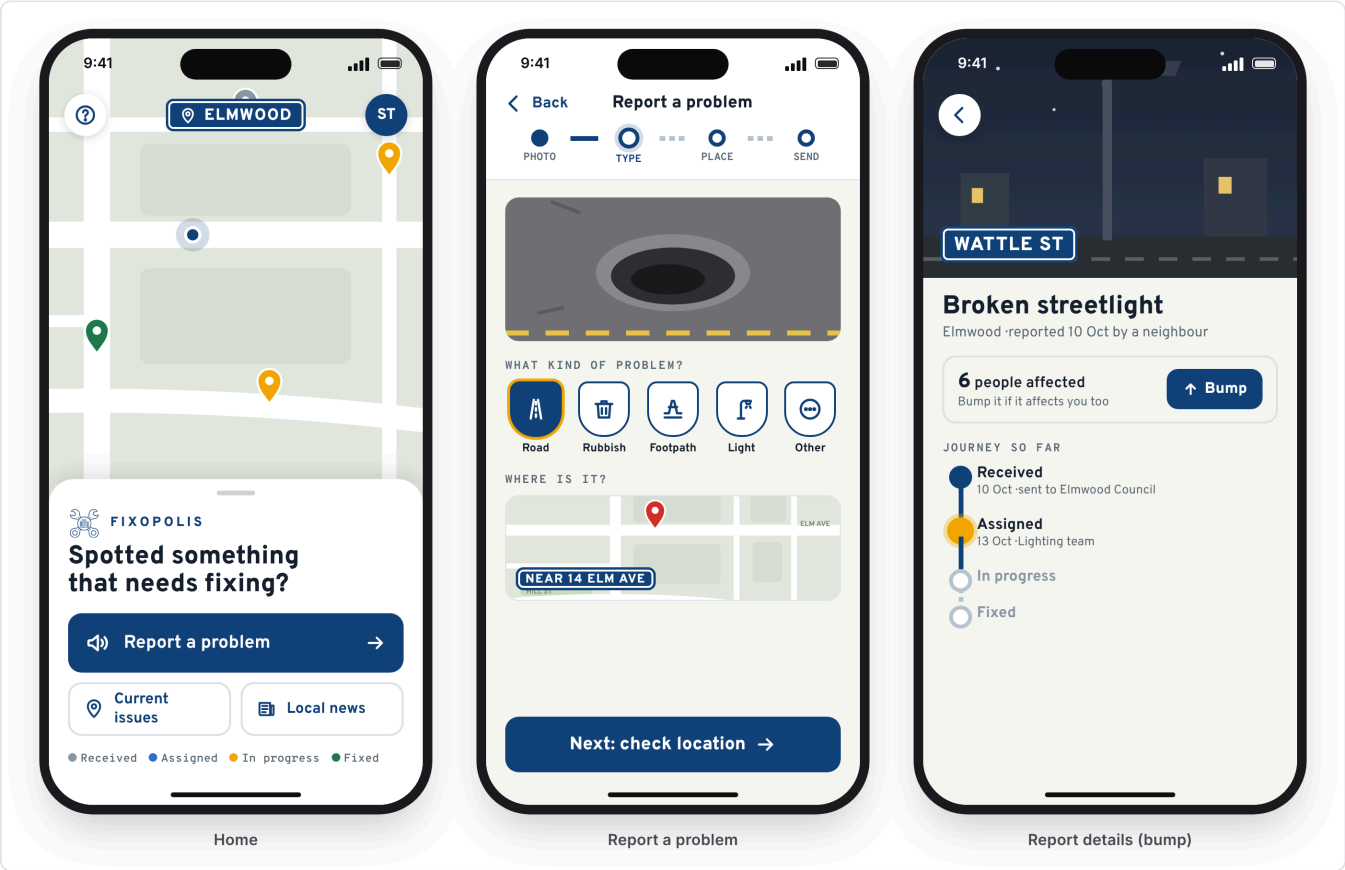
Palette Asphalt #22262A · Hi-vis yellow #FFC72C · Concrete #ECEBE6 · Signal orange #F2661B

Strengths Strongest character of the five; very high contrast; large tap targets.

Risks Feels "roadworks only" for rubbish or streetlights; long uppercase condensed text is slower to read.

Direction B: Wayfinding

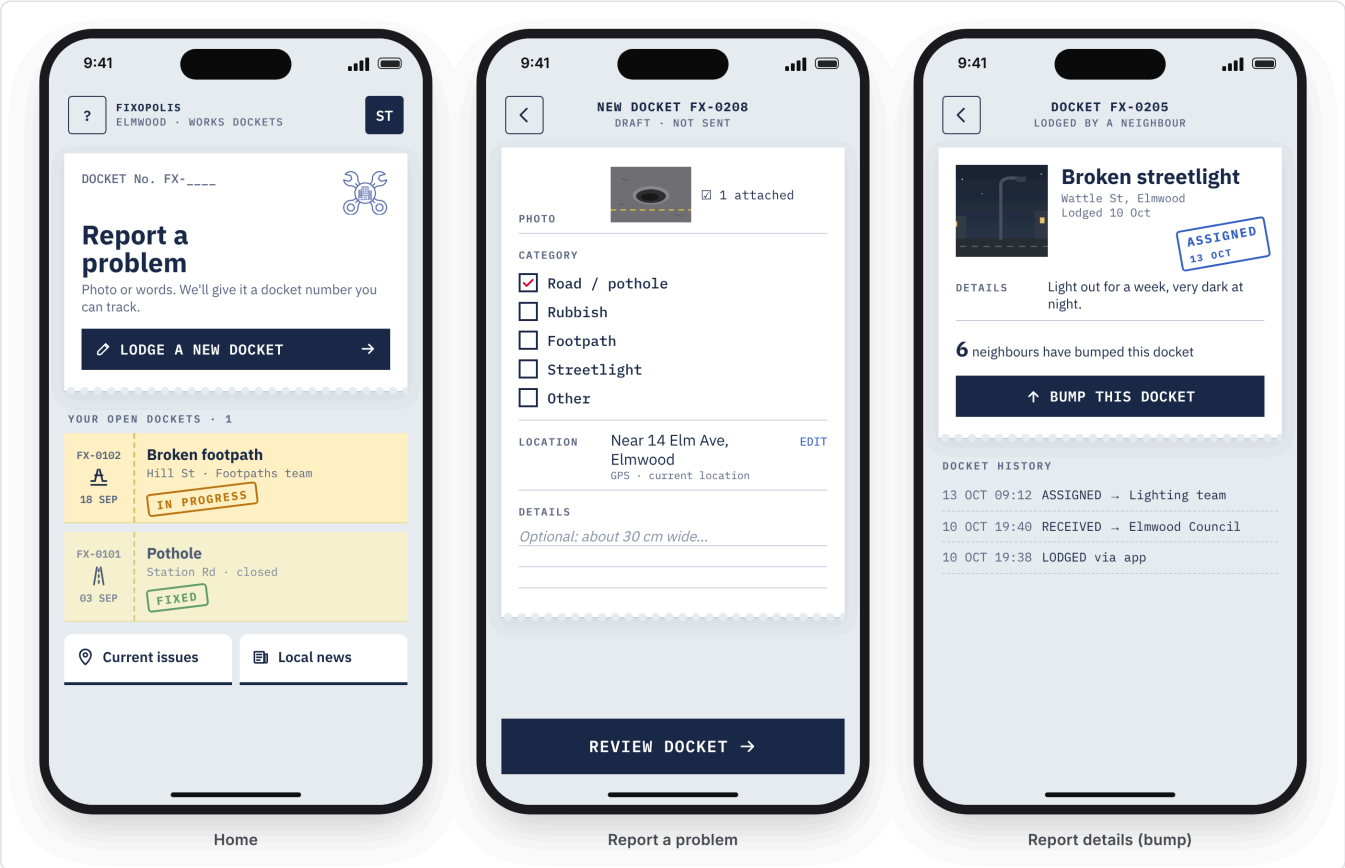
The city's own street signage. The map is the home screen, streets are labelled with street-name blades, and progress is a transit line.



Why it fits	Every report is about a place, so location leads. Signage is understood everywhere (Universality criterion). The transit line shows the steps still to come as well as those done, which speaks to the 61.9% who don't understand what their council is doing.
Type	Overpass (based on Highway Gothic, the lettering used on road signs) · Overpass Mono (labels)
Palette	Sign blue #12417A · Paper #F6F7F4 · Amber #F2A900 · Map green #E4EADF
Strengths	Nearby issues visible at a glance; progress shows what happens next, not just what has happened.
Risks	"Report a problem" shares Home with the map; most build effort (new home layout with a bottom sheet).

Direction C: Council Docket

Every report is a works docket: a reference number, rubber-stamp statuses, perforated slips and a timestamped docket history.



Why it fits Makes the council's paperwork visible, which targets the transparency and accountability gap raised in interviews (Fiona's and Artem's interviewees) and by the FixMyStreet reviewer. Strong real-world metaphor (H2).

Type IBM Plex Mono (reference numbers, stamps, labels) · IBM Plex Sans (body)

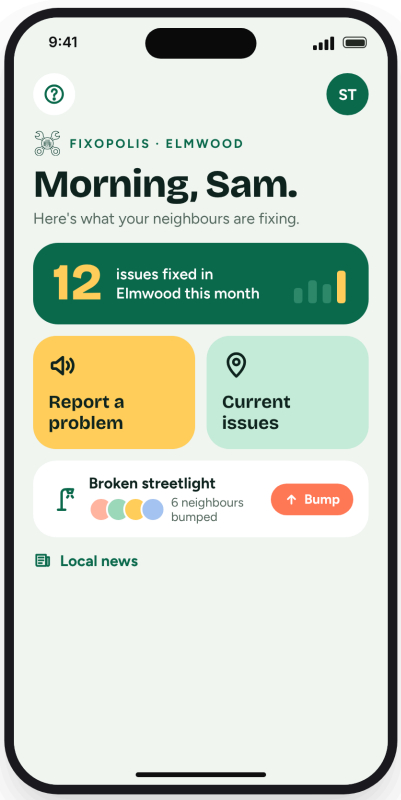
Palette Ballpoint navy #1C2B4A · Desk #E6EBF0 · Carbon copy #FFF4C4 · Stamp red #C8102E

Strengths Memorable; reference numbers are useful if a resident contacts the council.

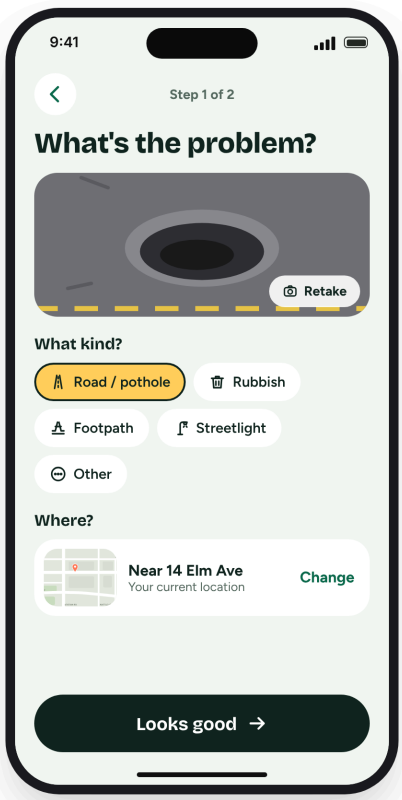
Risks A paperwork look could remind people of the bureaucracy that puts them off reporting; monospace text is denser.

Direction D: Neighbourhood

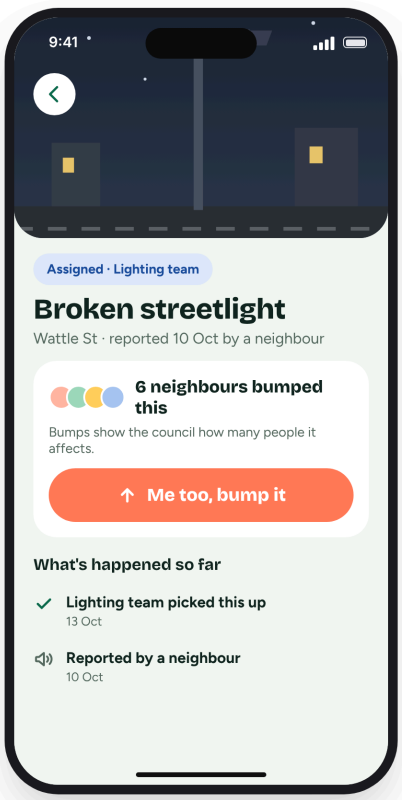
Community first: friendly type, warm colours, neighbours' avatars and how much has been fixed nearby.



Home



Report a problem



Report details (bump)

Why it fits Self-efficacy was the lowest survey item (mean 2.07). Showing collective action and results ("12 issues fixed this month") speaks to feeling powerless, and Olivia's interviewee wanted to see other reports being fixed. Bumping is the most prominent of the five.

Type Bricolage Grotesque (headings) · Figtree (body)

Palette Deep green #0E6B4F · Mint #C6EBD8 · Sun #FFD15C · Coral #FF7A59

Strengths Most welcoming for non-reporters; supports Task 3 strongly.

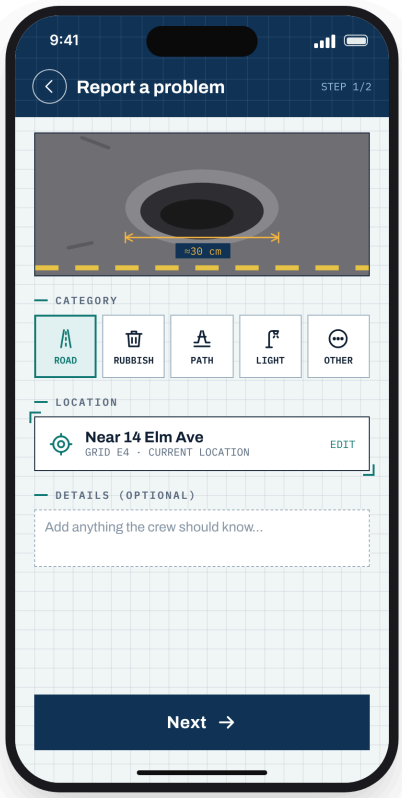
Risks Playful tone may read as less official; the community stat on Home is new content (from Part I "expected information").

Direction E: Blueprint

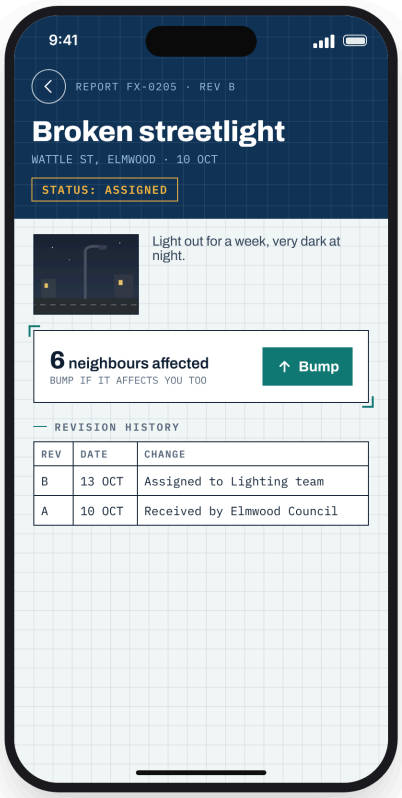
Engineering drawings: blueprint grid, the logo's line art, title blocks, dimension lines, and a revision table for status updates.



Home



Report a problem



Report details (bump)

Why it fits	Carries the logo's line-icon style across the whole app. A revision history is a precise real-world metaphor for status updates (H2).
Type	Archivo (headings, body) · IBM Plex Mono (annotations)
Palette	Blueprint navy #0F3557 · Grid paper #F3F6F9 · Logo teal #0F7C74 · Amber #F5B841
Strengths	Most consistent with the logo; calm and distinctive.
Risks	Technical feel may put off some users; small monospace capital labels need care for legibility.

Sources: INFO3315 Project Description 2026; group reference doc (Part I sections, PACT, interviews, survey results); Heuristic Evaluations Workbook; think-aloud and cognitive walkthrough notes; Lectures 6–8; Worksheets W7–W8.